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MONITORING DEVICE, MONITORING METHOD, AND NON-TRANSITORY COMPUTER READABLE MEDIUM

Abstract

A monitoring device (**100**) includes a communication connection unit (**124**). The communication connection unit (**124**) repeatedly executes connection to an unauthorized AP when the unauthorized AP exists in its surroundings, in order to fill simultaneous connection slots of the unauthorized AP, using each device identification information of one or more pieces of device identification information generated not to overlap each other. When connection to the unauthorized AP fails, the communication connection unit (**124**) connects to the unauthorized AP using device identification information of a damaged CL connected to the unauthorized AP, so as to disconnect communication connection between the unauthorized AP and the damaged CL, and connects to the unauthorized AP using other device identification information, in order to fill vacant simultaneous connection slots of the unauthorized AP that have become available due to disconnection of the communication connection.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION [0001] This application is a Continuation of PCT International Application No. PCT/JP2023/001230, filed on Jan. 17, 2023, which is hereby expressly incorporated by reference into the present application.

TECHNICAL FIELD

[0002] The present disclosure relates to a monitoring device, monitoring method, and monitoring program.

BACKGROUND ART

[0003] The Wi-Fi (registered trademark) security protocol including up to the security protocol WPA2 (Wi-Fi (registered trademark) Protected Access 2) had a vulnerability where a third party could forcibly disconnect an already established communication using a deauthentication frame, known as a DOS (Denial of Service). In view of this vulnerability, in the security protocol WPA3, countermeasures using a mechanism called PMF (Protected Management Frames) are implemented mandatorily.

[0004] Meanwhile, in WIPS (Wireless Intrusion Prevention System), the vulnerability related to the forced disconnection of communication by the deauthentication frame has been used for the purpose of blocking communication of unauthorized devices (including both access points and clients). Fundamental countermeasures against unauthorized devices include physical elimination of the unauthorized devices or cutting off the power supply to the unauthorized devices. However, it is not possible to immediately implement these fundamental countermeasures after detecting unauthorized communication. Therefore, to mitigate damage occurring between the detection of unauthorized communication and the implementation of fundamental countermeasures, forced disconnection of communication using deauthentication frames was employed. However, with the spread of WPA3, forced disconnection of communication using deauthentication frames will no longer function. Therefore, a different method that does not rely on deauthentication frames is needed to block communication of unauthorized devices.

[0005] Patent Literature 1 discloses a technology in which, when an unauthorized client is detected in a PMF environment, a WIPS monitoring device impersonates the MAC (Medium Access Control) address of the unauthorized client to connect to authorized access points, thereby updating an encryption key for encrypting communication frames related to the MAC address. As a result of the encryption key being updated, the unauthorized client will not be able to communicate with authorized access points because the unauthorized client does not know the new encryption key.

CITATION LIST

Patent Literature

[0006] Patent Literature 1: Japanese Patent Publication No. 2018-511282

SUMMARY OF INVENTION

Technical Problem

[0007] Patent Literature 1 has a problem of not disclosing countermeasures for a case where an unauthorized access point is installed, and a problem of not disclosing countermeasures to make it difficult to reconnect after a communication connection is disconnected.

[0008] The present disclosure aims to provide countermeasures for a case where an unauthorized access point is installed in a PMF-enabled Wi-Fi (registered trademark) network so as to prevent easy reconnection after a communication connection is disconnected.

Solution to Problem

[0009] A monitoring device according to the present disclosure includes [0010] a communication connection unit [0011] to repeatedly execute connection to an unauthorized access point when the unauthorized access point exists within a surrounding device group, in order to fill simultaneous connection slots of the unauthorized access point, using each device identification information from a device identification information group consisting of one or more pieces of device identification information generated not to overlap each other; [0012] to connect to the unauthorized access point when the connection to the unauthorized access point fails, using device identification information of a damaged client being a communication device connected to the unauthorized access point, so as to disconnect communication connection between the unauthorized access point and the damaged client; and [0013] to connect to the unauthorized access point using device identification information different from any device identification information in the device identification information group, in order to fill the simultaneous connection slots of the unauthorized access point that have become available due to disconnection of the communication connection, [0014] the surrounding device group consisting of one or more devices that exist around the monitoring device and execute wireless communication, [0015] each device identification information in the device identification information group being different from the device identification information of the damaged client.

Advantageous Effects of Invention

[0016] According to the present disclosure, simultaneous connection slots of an unauthorized access point are filled after disconnecting the communication connection between the unauthorized access point and a damaged client. Therefore, the damaged client cannot easily reconnect to the unauthorized access point after the communication connection is disconnected. Moreover, the present disclosure functions even in a PMF-enabled Wi-Fi (registered trademark) network. Therefore, according to the present disclosure, it is possible to provide a countermeasure in a case where an unauthorized access point is installed in a PMF-enabled Wi-Fi (registered trademark) network, in order to prevent easy reconnection after the communication connection is disconnected.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0017] FIG. 1 is a diagram describing a wireless communication network according to Embodiment 1.

[0018] FIG. 2 is a diagram showing a configuration example of a monitoring device **100** according to Embodiment 1.

[0019] FIG. 3 is a diagram showing a hardware configuration example of the monitoring device **100** according to Embodiment 1.

[0020] FIG. 4 is a flowchart showing the operation of the monitoring device **100** according to Embodiment 1.

[0021] FIG. 5 is a diagram showing a hardware configuration example of a monitoring device **100** according to a modification of Embodiment 1.

[0022] FIG. 6 is a diagram showing a configuration example of a monitoring device **100** according to Embodiment 2.

[0023] FIG. 7 is a flowchart showing the operation of the monitoring device **100** according to Embodiment 2.

[0024] FIG. 8 is a flowchart showing the operation of a monitoring device **100** according to a

modification of Embodiment 2.

[0025] FIG. **9** is a diagram showing a configuration example of a monitoring device **100** according to Embodiment 3.

[0026] FIG. **10** is a flowchart showing the operation of the monitoring device **100** according to Embodiment 3.

[0027] FIG. **11** is a diagram showing a configuration example of a monitoring device **100** according to Embodiment 4.

[0028] FIG. **12** is a flowchart showing the operation of the monitoring device **100** according to Embodiment 4.

[0029] FIG. **13** is a diagram showing a configuration example of a monitoring device **100** according to a modification of Embodiment 4.

[0030] FIG. **14** is a flowchart showing the operation of the monitoring device **100** according to the modification of Embodiment 4.

[0031] FIG. **15** is a diagram showing a configuration example of a monitoring device **100** according to Embodiment 5.

[0032] FIG. **16** is a flowchart showing the operation of the monitoring device **100** according to Embodiment 5.

[0033] FIG. **17** is a diagram describing a wireless communication network according to the modification of Embodiment 4 and a modification of Embodiment 5.

DESCRIPTION OF EMBODIMENTS

[0034] In the description and drawings of the Embodiment, the same reference numerals are assigned to the same elements and corresponding elements. The description of elements with the same reference numerals is omitted or simplified as appropriate. Arrows in diagrams mainly indicate flows of data or flows of processing. Also, the term “unit” may be appropriately replaced with “circuit”, “stage”, “procedure”, “processing”, or “circuitry”.

Embodiment 1

[0035] Hereinafter, the present Embodiment will be described in detail with reference to the drawings.

[0036] In this Embodiment, the countermeasure against the unauthorized access point is a countermeasure in a situation where, as shown in FIG. **1**, an unauthorized AP (access point) **10** has intruded into a wireless communication network and a communication connection has already been established between the unauthorized AP **10** and authorized CLs (clients) **20**. The wireless communication network is, as a specific example, a Wi-Fi (registered trademark) network where PMF (Protected Management Frames) is enabled. The authorized CL **20** is a communication device, specifically a communication terminal such as a smartphone and a PC (Personal Computer). Here, among the authorized CLs **20**, a CL that has already established a communication connection with the unauthorized AP **10** is particularly called a damaged CL **21**. The damaged CL **21** is a communication device connected to the unauthorized AP **10**. Also, the communication between the damaged CL **21** and the unauthorized AP **10** is called unauthorized communication. In unauthorized communication, PMF may be enabled, and there is a risk that, for example, the information of the damaged CL **21** may be stolen by the unauthorized AP **10**, or the damaged CL **21** may be forced to download malware from the unauthorized AP **10**.

[0037] In this Embodiment, a countermeasure that a monitoring device **100** can implement in the aforementioned situation is shown. The monitoring device **100** may be at least a part of an authorized AP **30**, or may be a device installed independently of the authorized AP **30**. Also, when the monitoring device **100** is at least a part of the authorized AP **30**, the authorized AP **30** may be equipped with multiple wireless devices, and the operation as an AP and the countermeasure shown in this Embodiment may be implemented by distributing the operation and the countermeasure in the multiple wireless devices.

Description of Configuration

[0038] FIG. 2 shows an example configuration of the monitoring device **100** according to Embodiment 1. The monitoring device **100**, as shown in FIG. 2, includes a communication unit **110**, a control unit **120**, and a storage unit **130**. The monitoring device **100** is also referred to as a wireless intrusion prevention device. In addition, when the monitoring device **100** is implemented in the authorized AP **30**, the monitoring device **100** may further include a communication processing unit that realizes functions as an access point.

[0039] The communication unit **110** includes an antenna for wireless communication and has the function of sending and receiving data via wireless communication to and from other devices.

[0040] The control unit **120** includes a communication monitoring unit **121**, an information analysis unit **122**, an unauthorized device judging unit **123**, a communication connection unit **124**, and a MAC (Medium Access Control) address generation unit **125**.

[0041] The communication monitoring unit **121** acquires communication frames from each device in a surrounding device group. Here, the surrounding device group consists of one or more devices that exist around the monitoring device **100** and execute wireless communication. The surrounding device group includes at least any one of an unauthorized AP **10**, an authorized CL **20**, and a damaged CL **21**.

[0042] The information analysis unit **122** analyzes the communication frames acquired by the communication monitoring unit **121**.

[0043] The unauthorized device judging unit **123** judges whether each device in the surrounding device group is an unauthorized device. As a specific example, the unauthorized device judging unit **123** judges whether an unauthorized AP **10** exists within the surrounding device group based on a communication frame sent by each device in the surrounding device group.

[0044] The communication connection unit **124** executes the process to connect with other devices via wireless communication. In this specification, the term “connection” primarily refers to association via wireless communication.

[0045] Below, a specific example of the process of the communication connection unit **124** is described. In this example, PMF may be enabled in wireless communication between the unauthorized AP **10** and the damaged CL **21**.

[0046] First, the communication connection unit **124** repeatedly executes connection to an unauthorized AP **10** when the unauthorized AP **10** exists within the surrounding device group, in order to fill the simultaneous connection slots of the unauthorized AP **10**, using each device identification information from a device identification information group. Here, the device identification information group consists of one or more pieces of device identification information generated not to overlap each other. Also, each device identification information in the device identification information group is different from device identification information of the damaged CL **21**. Each device identification information is, as a specific example, a MAC address.

[0047] Next, the communication connection unit **124** connects to the unauthorized AP **10** when the connection to the unauthorized AP **10** fails, using the device identification information of the damaged CL **21**, so as to disconnect the communication connection between the unauthorized AP **10** and the damaged CL **21**.

[0048] Then, the communication connection unit **124** connects to the unauthorized AP **10** using device identification information different from any device identification information in the device identification information group, in order to fill the simultaneous connection slots of the unauthorized AP **10** that have become available due to disconnection of the communication connection.

[0049] The MAC address generation unit **125** generates MAC addresses as needed.

[0050] The storage unit **130** stores communication frame information **131**, unauthorized device information **132**, damaged device information **133**, and a used MAC address table **134**.

[0051] The communication frame information **131** consists of communication frames sent by the individual devices in the surrounding device group.

[0052] The unauthorized device information **132** consists of device information of the individual unauthorized devices. An unauthorized device is a device that is not an authorized device. The unauthorized device may be a device not registered in a list indicating authorized devices, or a device registered in a list indicating unauthorized devices.

[0053] The damaged device information **133** consists of device information of the individual damaged devices. A damaged device is a device that is connected to an unauthorized device.

[0054] The used MAC address table **134** is table data indicating a list of MAC addresses that have been used.

[0055] FIG. **3** shows a hardware configuration example of the monitoring device **100** according to this Embodiment. The monitoring device **100** consists of a computer. The monitoring device **100** may consist of multiple computers. Additionally, the monitoring device **100** may be configured to incorporate a computer.

[0056] The monitoring device **100** is a computer equipped with hardware such as a processor **51**, a memory **52**, an auxiliary storage device **53**, an input/output IF (Interface) **54**, and a communication device **55**, as shown in this figure. These hardware components are appropriately connected via a signal line **59**.

[0057] The processor **51** is an IC (Integrated Circuit) that performs arithmetic processing and controls the hardware provided to the computer. The processor **51** is, for example, a CPU (Central Processing Unit), DSP (Digital Signal Processor), or a GPU (Graphics Processing Unit).

[0058] The monitoring device **100** may include a plurality of processors as an alternative to the processor **51**. The plurality of processors share roles of the processor **51**.

[0059] The memory **52** is typically a volatile storage device, constituting the storage unit **130**, and is, for example, a RAM (Random Access Memory). The memory **52** is also called a main storage device or main memory. Data stored in the memory **52** is saved in the auxiliary storage device **53** as needed.

[0060] The auxiliary storage device **53** is typically a non-volatile storage device, also called storage, and is, for example, a ROM (Read Only Memory), an HDD (Hard Disk Drive), or a flash memory. Data stored in the auxiliary storage device **53** is loaded into the memory **52** as needed.

[0061] The memory **52** and the auxiliary storage device **53** may be configured integrally.

[0062] The input/output IF **54** is a port to which input and output devices are connected. The input/output IF **54** is, for example, a USB (Universal Serial Bus) terminal. The input device is, for example, a keyboard and a mouse. The output device is, for example, a display.

[0063] The communication device **55** is a receiver/transmitter. The communication device **55**, as a specific example, is a communication chip or NIC (Network Interface Card).

[0064] Each part of the monitoring device **100** may appropriately use the input/output IF **54** and the communication device **55** when communicating with other devices, etc.

[0065] The auxiliary storage device **53** stores a monitoring program. The monitoring program is a program that enables the function of each part of the monitoring device **100** to be realized on a computer. By the monitoring program stored in the auxiliary storage device **53** being loaded into the memory **52** and executed by the processor **51**, the functions of this Embodiment are implemented. The function of each part of the monitoring device **100** is implemented by software.

[0066] Data used when executing the monitoring program and data obtained by executing the monitoring program, etc., are appropriately stored in the storage device. Each part of the monitoring device **100** appropriately utilizes the storage device. The storage device consists of at least one of the memory **52**, the auxiliary storage device **53**, a register within the processor **51**, and a cache memory within the processor **51**. It should be noted that the term data and the term information may sometimes have equivalent meanings. The storage device may be independent of the computer.

[0067] The functions of the memory **52** and auxiliary storage device **53** may be implemented by other storage devices.

[0068] The monitoring program may be recorded on a computer readable non-volatile recording medium. As a specific example, the non-volatile recording medium is an optical disc or flash memory. The monitoring program may be provided as a program product.

Description of Operation

[0069] The operation procedure of the monitoring device **100** corresponds to a monitoring method. Additionally, a program that realizes the operation of the monitoring device **100** corresponds to the monitoring program.

[0070] FIG. **4** is a flowchart showing an example of the operation of the monitoring device **100** according to Embodiment 1. FIG. **4** is used to describe the operation of the monitoring device **100** according to Embodiment 1.

(Step S101)

[0071] First, the communication unit **110** receives communication radio waves from each device in the surrounding device group using an antenna, demodulates the received radio waves, and performs analog-digital conversion on the demodulated results to obtain a digital signal.

[0072] Next, the communication unit **110** sends the obtained digital signal to the communication monitoring unit **121**.

[0073] Next, the communication monitoring unit **121** interprets the received digital signal as a communication frame, thereby acquiring a communication frame sent and received by each device in the surrounding device group. The communication monitoring unit **121** stores the acquired communication frame in the storage unit **130** as part of the communication frame information **131**.

(Step S102)

[0074] First, the information analysis unit **122** analyzes the communication frame information **131** to acquire information on each device in the surrounding device group. As a specific example, the information of the unauthorized AP **10** is information obtained from a beacon frame that the unauthorized AP **10** sends periodically, indicating at least one of the unauthorized AP **10**'s BSSID (Basic Service Set Identifier), ESSID (Extended Service Set Identifier), the channel in use (communication frequency), PMF enabled/disabled, authentication method, encryption method, and MAC address (usually the same as BSSID). Additionally, the information of the authorized CL **20** or the damaged CL **21** is information obtained from a communication frame addressed to an AP or from a probe request frame sent to search for an AP, indicating at least any one of the MAC address of the authorized CL **20** or the damaged CL **21**, and the ESSID and BSSID of the AP to which the authorized CL **20** or the damaged CL **21** is connected (or searching).

[0075] Next, the unauthorized device judging unit **123** judges whether an unauthorized AP **10** exists in the surrounding device group based on the information acquired by the information analysis unit **122**. The method to judge whether an unauthorized AP **10** exists includes, as a specific example, judging that the unauthorized AP **10** exists if a device matching a device on the list of predetermined unauthorized device information is included in the surrounding device group, judging that an unauthorized AP **10** exists if a device not shown in the list of the predetermined authorized device information is included in the surrounding device group, judging that an unauthorized AP **10** exists if a device with the same device information as the monitoring device **100** is included in the surrounding device group, and judging that an unauthorized AP **10** exists if there is an unexpected response (or no response) when accessing an AP indicated by the acquired information via the wired network to which the authorized AP **30** is connected.

(Step S103)

[0076] If an unauthorized AP **10** exists in the surrounding device group, the monitoring device **100** proceeds to step S104. If an unauthorized AP **10** does not exist in the surrounding device group, the monitoring device **100** returns to step S101.

(Step S104)

[0077] The unauthorized device judging unit **123** stores the device information of the unauthorized AP **10** detected in step S102 to the storage unit **130** as part of the unauthorized device information

132. Here, the device information corresponding to each device is assumed to include a countermeasure implementation flag corresponding to each device. The countermeasure implementation flag corresponding to each device is a flag to indicate whether countermeasures have been implemented for each device. When a countermeasure implementation flag corresponding to a certain device is set, this indicates that countermeasures have been implemented for that device. It is possible that an unauthorized AP **10**, for which countermeasures have been implemented once, may attempt to attack again after rebooting. Therefore, implementation may be performed such that the countermeasure implementation flag corresponding to each device is reset (to a state indicating countermeasures not implemented) after a specified time has elapsed since the flag was set.

[0078] Regarding the unauthorized device information **132**, each device information may be deleted after a specified time has elapsed since the device information is added to the unauthorized device information **132**, or the unauthorized device information **132** may be reset each time step **S104** is executed. Additionally, the unauthorized device judging unit **123** may fix an upper limit on the number of unauthorized APs **10**, and if the number of unauthorized APs **10** exceeds the fixed upper limit, the unauthorized device judging unit **123** may delete the oldest device information from the unauthorized device information **132** and then add new device information to the unauthorized device information **132**.

(Step **S105**)

[0079] If there is no device information without a countermeasure implementation flag being set in the unauthorized device information **132** (that is, if a countermeasure implementation flag is set for device information of all the unauthorized APs **10** indicated by the unauthorized device information **132**), the monitoring device **100** returns to step **S101**. Otherwise, the monitoring device **100** proceeds to step **S106**.

(Step **S106**)

[0080] The unauthorized device judging unit **123** selects one piece of device information not having a countermeasure implementation flag being set, from the device information of the unauthorized AP **10** indicated by the unauthorized device information **132**. Hereafter, the unauthorized AP **10** selected in step **S106** is referred to as the selected unauthorized AP.

(Step **S107**)

[0081] First, the unauthorized device judging unit **123** requires the information analysis unit **122** to acquire the device information of each damaged CL **21** connected to the selected unauthorized AP.

[0082] Next, the information analysis unit **122** acquires the device information of each damaged CL **21** connected to the selected unauthorized AP and sends the acquired device information to the unauthorized device judging unit **123**.

(Step **S108**)

[0083] The unauthorized device judging unit **123** stores the device information, sent by the information analysis unit **122**, of each damaged CL **21** connected to the selected unauthorized AP, to the storage unit **130** as part of the damaged device information **133**. Here, the device information corresponding to each device is assumed to include a countermeasure implementation flag corresponding to that device.

[0084] Regarding the damaged device information **133**, each device information may be deleted after a specified time has elapsed since the device information is added to the damaged device information **133**, or the damaged device information **133** may be reset each time step **S108** is executed. Additionally, the unauthorized device judging unit **123** may fix an upper limit on the number of damaged CLs **21**, and if the number of damaged CLs **21** exceeds the fixed upper limit, the unauthorized device judging unit **123** may delete the oldest device information from the damaged device information **133** and then add new device information to the damaged device information **133**. However, if adopting, among the three methods mentioned above, the two methods other than the method of resetting the damaged device information **133** each time step

S108 is executed, there may be multiple pairs of unauthorized AP **10** and damaged CL **21**.

Therefore, it is necessary to store information indicating the unauthorized AP **10** to which each damaged CL **21** is connected (for example, the BSSID of the unauthorized AP **10**), such that the information is linked to each device information.

(Step **S109**)

[0085] The MAC address generation unit **125** generates one MAC address. The MAC address generation unit **125** may generate the MAC address using a random number, or by incrementing the value of the MAC address indicated by the damaged device information **133**. The MAC address generation unit **125** refers to the damaged device information **133** and the used MAC address table to confirm that the generated MAC address is different from any MAC address in a confirmation target MAC address group. The confirmation target MAC address group consists of the MAC addresses of the individual damaged CLs **21** and the MAC addresses stored in the used MAC address table.

[0086] The MAC address generation unit **125**, upon confirming that the generated MAC address is different from any MAC address in the confirmation target MAC address group, adds the generated MAC address to the used MAC address table **134** and sends data indicating the generated MAC address to the communication connection unit **124**. If the generated MAC address overlaps any MAC address in the confirmation target MAC address group, the MAC address generation unit **125** regenerates the MAC address and then reconfirms that the newly generated MAC address is different from any MAC address in the confirmation target MAC address group.

[0087] Here, if the generated MAC addresses are continuously stored permanently in the used MAC address table **134**, there is a possibility of falling into an endless loop of MAC address generation and confirmation failure due to MAC address exhaustion. Therefore, each MAC address stored in the used MAC address table **134** may be deleted after a specified time has elapsed since it is added to the used MAC address table **134**, or the used MAC address table **134** may be reset each time step **S106** is executed. Additionally, the MAC address generation unit **125** may fix an upper limit on the number of MAC addresses that can be stored, and if the number of MAC addresses exceeds the fixed upper limit, the MAC address generation unit **125** may delete the oldest MAC address from the used MAC address table **134** and then store a new MAC address in the used MAC address table **134**.

(Step **S110**)

[0088] The communication connection unit **124** generates a communication frame necessary for a connection process as a digital signal. At this time, the communication connection unit **124** sets the MAC address generated in step **S109**, as the transmission source MAC address.

(Step **S111**)

[0089] The communication connection unit **124** executes the connection process by sending an authentication request frame and an association request frame, etc., to the unauthorized AP **10** via the communication unit **110**. The communication connection unit **124** confirms that the connection process is executed normally by checking the response from the communication unit **110**. If encryption of communication is set, the communication connection unit **124** may proceed with the connection process up to sharing of the encryption key.

(Step **S112**)

[0090] If the monitoring device **100** can connect normally to the unauthorized AP **10**, the monitoring device **100** returns to step **S109**.

[0091] If the monitoring device **100** cannot connect normally to the unauthorized AP **10**, it is determined that the maximum number of simultaneous connections to the unauthorized AP **10** has been reached, and the monitoring device **100** proceeds to step **S113**. If the monitoring device **100** cannot connect normally to the unauthorized AP **10**, specifically, this is when an error is returned, or when the monitoring device **100** cannot connect normally to the unauthorized AP **10** because communication timeout occurs, etc.

(Step S113)

[0092] If there is no device information without a countermeasure implementation flag being set in the damaged device information **133** (i.e., if the countermeasure implementation flag is set for device information of all the damaged CL **21** indicated by the damaged device information **133**), the unauthorized device judging unit **123** sets the countermeasure implementation flag in the device information of the selected unauthorized AP in the unauthorized device information **132**, judging that countermeasures have been implemented for the selected unauthorized AP, and then the monitoring device **100** returns to step **S105**. Otherwise, the monitoring device **100** proceeds to step **S114**.

(Step S114)

[0093] The communication connection unit **124** generates a communication frame necessary for the connection process as a digital signal. At this time, the communication connection unit **124** refers to the damaged device information **133** to select one damaged CL **21** for which countermeasures have not been implemented, and sets the MAC address of the selected damaged CL **21**, as the transmission source MAC address of the communication frame. Here, the damaged CL **21** for which countermeasures have not been implemented corresponds to the damaged CL **21** among the device information included in the damaged device information **133**, where the countermeasure implementation flag is not set. Hereinafter, the damaged CL **21** selected in step **S114** will be referred to as the first selected damaged CL.

(Step S115)

[0094] The communication connection unit **124** executes the connection process by sending an authentication request frame and an association request frame, etc., to the unauthorized AP **10** via the communication unit **110**. The communication connection unit **124** confirms that the connection process has been executed normally by checking the response from the communication unit **110**. If encryption of communication is set, the communication connection unit **124** proceeds with the connection process up to the sharing of the encryption key.

[0095] Once the connection process with the unauthorized AP **10** is completed, the communication connection unit **124** sends a deauthentication frame in which the transmission source MAC address is set as the MAC address of the first selected damaged CL, to the unauthorized AP **10**, thereby disconnecting the communication connection between the unauthorized AP **10** and the first selected damaged CL. Here, in a case where communication encryption is set, since the monitoring device **100** can obtain the communication encryption key by proceeding with the sharing process of the encryption key, the unauthorized AP **10** can be made to process the deauthentication frame normally even if PMF is enabled.

[0096] In cases where communication encryption is set, depending on the implementation of the unauthorized AP **10**, it may be possible to disconnect the communication connection between the unauthorized AP **10** and the first selected damaged CL by simply sending a deauthentication frame with the transmission source MAC address being set as the MAC address of the first selected damaged CL, to the unauthorized AP **10** during the connection process, without proceeding with the connection process up to the sharing of the encryption key.

[0097] Through the aforementioned process, unauthorized communication between the first selected damaged CL and the unauthorized AP **10** is disconnected. The communication connection unit **124** sets, in the damaged device information **133**, a countermeasure implementation flag in the device information of the first selected damaged CL in order to record that the unauthorized communication has been disconnected.

[0098] Meanwhile, the disconnection of the communication connection frees up one slot in the simultaneous connection capacity of the unauthorized AP **10**. To prevent the damaged CL **21** from reconnecting to the freed slot, the monitoring device **100** returns to step **S109** after executing step **S115**.

Explanation of the Effects of Embodiment 1

[0099] As described above, in this Embodiment, unauthorized communication between the damaged CL **21** and the unauthorized AP **10** is disconnected. Furthermore, since the monitoring device **100** fills the simultaneous connection slots of the unauthorized AP **10**, the damaged CL **21**, whose unauthorized communication has been disconnected, cannot reconnect to the unauthorized AP **10**. Therefore, according to this Embodiment, it is possible to mitigate damages such as the information of the damaged CL **21** being stolen by the unauthorized AP **10** or the damaged CL **21** being forced to download malware from the unauthorized AP **10**. Additionally, according to this Embodiment, it is possible to buy time needed until fundamental countermeasures such as the physical elimination of unauthorized devices or cutting off the power to unauthorized devices are implemented.

Other Configurations

<Modification 1>

[0100] FIG. **5** shows a hardware configuration example of a monitoring device **100** according to this modification.

[0101] The monitoring device **100**, instead of: the processor **51**; the processor **51** and the memory **52**; the processor **51** and the auxiliary storage device **53**; or the processor **51**, the memory **52**, and the auxiliary storage device **53**, includes a processing circuit **58**. The processing circuit **58** is hardware that realizes at least part of the components provided to the monitoring device **100**.

[0102] The processing circuit **58** may be dedicated hardware, or a processor that executes a program stored in the memory **52**.

[0103] When the processing circuit **58** is dedicated hardware, the processing circuit **58** may be, for example, a single circuit, a composite circuit, a programmed processor, a parallel-programmed processor, an ASIC (Application Specific Integrated Circuit), an FPGA (Field Programmable Gate Array), or a combination of these.

[0104] The monitoring device **100** may include a plurality of processing circuits as an alternative to the processing circuit **58**. The plurality of processing circuits share the role of the processing circuit **58**.

[0105] In the monitoring device **100**, some functions may be implemented by dedicated hardware, and the remaining functions may be implemented by software or firmware.

[0106] The processing circuit **58** may be implemented by hardware, software, firmware, or a combination of these, as a specific example.

[0107] The processor **51**, the memory **52**, the auxiliary storage device **53**, and the processing circuit **58** are collectively referred to as “processing circuitry”. In other words, the functions of the individual functional components of the monitoring device **100** are implemented by the processing circuitry.

[0108] A monitoring device **100** according to another Embodiment may also have the same configuration as this modification.

Embodiment 2

[0109] Below, the points differing from the aforementioned Embodiment will mainly be described with reference to the drawings.

[0110] In Embodiment 1, when a large number of the damaged CLs **21** exist, it may require a certain amount of time to implement countermeasures for all the damaged CLs **21**. Therefore, in Embodiment 2, a mode is shown where the damaged CL **21** is retreated to another channel as preparation on ahead before filling the simultaneous connection slots of the unauthorized AP **10**.

Description of Configuration

[0111] FIG. **6** shows a configuration example of a monitoring device **100** according to Embodiment 2.

[0112] A control unit **120** according to Embodiment 2, as shown in FIG. **6**, further includes a probe response generation unit **126** and a channel selection unit **127** when compared to the control unit **120** according to Embodiment 1.

[0113] The probe response generation unit **126** generates a communication frame including information to switch the channel of a damaged CL **21** to a channel selected by the channel selection unit **127**. The communication frame is, as a specific example, a beacon frame or a probe response frame.

[0114] As a specific example, the probe response generation unit **126** sends first transmission data to the damaged CL **21** so as to switch the channel of the damaged CL **21** to a first channel. The first transmission data is communication data including: a communication frame in which device identification information of the unauthorized AP **10** is set as source device identification information so that the monitoring device **100** is recognized as an unauthorized AP **10**; and a communication frame in which the first channel is set as a channel to switch to. The first channel is a channel where an unauthorized AP **10** does not exist. It should be noted that setting the device identification information of the unauthorized AP **10** as the source device identification information so that the monitoring device **100** is recognized as the unauthorized AP **10** corresponds to the monitoring device **100** impersonating the unauthorized AP **10**.

[0115] The channel selection unit **127** selects a channel where an unauthorized AP **10** does not exist, as a channel switching destination of the damaged CL **21**.

[0116] The storage unit **130** according to Embodiment 2, as shown in FIG. 6, further

[0117] stores a relation table **135** when compared to the storage unit **130** according to Embodiment 1.

[0118] The relation table **135** is table data indicating pairs each consisting of an individual damaged device and a channel where the damaged device exists, and is also called a “relation table of damaged devices and channels”.

Description of Operation

[0119] FIG. 7 is a flowchart showing an example of the operation of the monitoring device **100** according to Embodiment 2. Using FIG. 7, the operation of the monitoring device **100** according to Embodiment 2 is described.

(Step S201)

[0120] First, the channel selection unit **127** refers to damaged device information **133** and selects one damaged CL **21** for which countermeasures have not been implemented. Hereinafter, the damaged CL **21** selected in step S201 is referred to as a second selected damaged CL.

[0121] Next, the channel selection unit **127** refers to the relation table **135** and selects one channel as the destination channel to which the second selected damaged CL is to shift, where an unauthorized AP **10** does not exist and the number of related damaged CLs **21** is minimal. If there are multiple of such channels, the channel selection unit **127** may select a channel with the smallest channel number or use a random number to select the channel. Hereinafter, the channel selected in step S201 is referred to as a first selected channel. If an unauthorized AP **10** exists on all channels, there is no effect in shifting the second selected damaged CL to another channel, so the monitoring device **100** terminates the process of step S201 and proceeds to step S109.

[0122] Next, the channel selection unit **127** adds a pair of the second selected damaged CL and the first selected channel to the relation table **135**. Additionally, the channel selection unit **127** sends data indicating the pair of the second selected damaged CL and the first selected channel to the probe response generation unit **126**.

(Step S202)

[0123] CSA (Channel Switch Announcement) is information sent to the CL when switching the channel through which the AP communicates. CSA is included in the beacon frame or the probe response frame, etc.

[0124] The probe response generation unit **126** sets the MAC address of the sender as the MAC address of the unauthorized AP **10**, and sends a probe response frame in which the first selected channel is set as the destination channel for switching in the CSA, to the second damaged CL via the communication unit **110**. Changing the content of the CSA corresponds to tampering with the

CSA.

[0125] After the transmission is completed, the probe response generation unit **126** sets a countermeasure implementation flag in the device information of the second damaged CL in the damaged device information **133** to record that the channel has been switched. When the damaged CL **21** is shifted to a channel different from the unauthorized AP **10**, the unauthorized AP **10** can no longer communicate with the damaged CL **21**, so it is permissible to share the flag indicating the channel switching and the countermeasure implementation flag. Note that channel switching corresponds to channel hopping.

(Step **S203**)

[0126] If there is no device information without a countermeasure implementation flag being set in the damaged device information **133** (i.e., if the countermeasure implementation flag is set for device information of all the damaged CLs **21** indicated by the damaged device information **133**), the monitoring device **100** proceeds to step **S109** to fill the simultaneous connection slots of the unauthorized AP **10**. Otherwise, the monitoring device **100** proceeds to step **S201**.

Explanation of the Effects of Embodiment 2

[0127] As described above, in this Embodiment, by switching the channel before filling the simultaneous connection slots of the unauthorized AP **10**, unauthorized communication between the damaged CL **21** and the unauthorized AP **10** is disconnected.

[0128] Furthermore, by filling the simultaneous connection slots of the unauthorized AP **10**, the disconnected damaged CL **21** will not reconnect to the unauthorized AP **10**. Additionally, according to this Embodiment, effects similar to those of Embodiment 1 can be obtained.

[0129] Moreover, in Embodiment 1, since the process of filling the simultaneous connection slots of the unauthorized AP **10** is executed on ahead, it requires time to disconnect the unauthorized communication between the damaged CL **21** and the unauthorized AP **10**. However, according to this Embodiment, since it is possible to disconnect the unauthorized communication before executing the process of filling the simultaneous connection slots of the unauthorized AP **10**, the damage can be mitigated more compared to Embodiment 1.

Other Configurations

<Modification 2>

[0130] In Embodiment 2, the damaged CL **21** is shifted to a channel determined randomly. If there is no authorized AP **30** at the shift destination, the shifted damaged CL **21** cannot communicate at all. Therefore, in this modification, assuming the presence of multiple wireless devices, authorized communication and countermeasures against the unauthorized AP **10** are implemented in parallel. Here, the authorized AP **30** may have multiple wireless devices, leading to the presence of multiple wireless devices. Each of the authorized AP **30** and the monitoring device **100**, being independent of the authorized AP **30**, may have one or more wireless devices, leading to the presence of multiple wireless devices.

[0131] This modification indicates a mode where, after executing the process of step **S203**, before executing the process of step **S109**, the damaged CL **21**, which has been shifted to another channel, is shifted back to the channel of the authorized AP **30**, so while authorized communication is conducted with one wireless device, the simultaneous connection slots of the unauthorized AP **10** are filled with another wireless device.

[0132] Below, the points different from Embodiment 2 are mainly described.

Description of Configuration

[0133] The configuration of a monitoring device **100** according to this modification is the same as the configuration of the monitoring device **100** according to Embodiment 2.

[0134] When the monitoring device **100** functions as an authorized AP **30**, a probe response generation unit **126** according to this modification sends second transmission data to a damaged CL **21** whose channel has been switched to a first channel, so as to switch the channel of the damaged CL **21** to a second channel. The second transmission data is communication data including: a

communication frame in which the device identification information of the unauthorized AP **10** is set as the source device identification information so that the monitoring device **100** is recognized as an unauthorized AP **10**; and a communication frame in which the second channel is set as a channel switching destination. The second channel is a channel where an unauthorized AP **10** does not exist. After that, the probe response generation unit **126** switches a channel of the monitoring device **100** to the second channel to establish authorized communication for the damaged CL **21**.

Description of Operation

[0135] FIG. **8** is a flowchart showing an example of the operation of the monitoring device **100** according to this modification. Using FIG. **8**, the operation of the monitoring device **100** is described.

(Step S221)

[0136] The channel selection unit **127** selects one channel where an unauthorized AP **10** does not exist, as the channel to resume authorized communication. Hereafter, the channel selected in step S221 is referred to as a second selected channel.

[0137] However, in this modification, assuming the possibility that the unauthorized AP **10** may perform channel shifting following the CSA sent in step S202, the channel selection unit **127** reacquires surrounding communication frames before selecting a channel, and confirms the channel where the unauthorized AP **10** exists based on the acquired communication frames.

(Step S222)

[0138] The probe response generation unit **126** sequentially sets the source MAC address as the MAC address of the unauthorized AP **10**, and sends a probe response frame in which the second selected channel is set as the channel to switch to in the CSA, to the damaged CL **21** indicated by the relation table **135**, via the communication unit **110**.

[0139] At the time of sending the probe response frame, the monitoring device **100** that sends the probe response frame needs to shift to a channel where the damaged CL **21**, being the destination of the probe response frame, exists.

(Step S223)

[0140] The probe response generation unit **126** shifts the channel of the authorized AP **30** to the second selected channel and sends the probe response frame to the damaged CL **21** as the authorized AP **30**. As a result, the damaged CL **21** can connect with the authorized AP **30** and resume authorized communication.

[0141] Meanwhile, the wireless device responsible for filling the simultaneous connection slots of the unauthorized AP **10** shifts its channel to the channel where an unauthorized AP **10** exists, and proceeds to step S109 to continue processing.

Explanation of the Effects of Modification 2

[0142] In Embodiment 2, the damaged CL **21** is shifted to a randomly determined channel. If an authorized AP **30** does not exist at the destination, the shifted damaged CL **21** cannot communicate at all. However, according to this modification, it is possible to simultaneously implement communication between the authorized AP **30** and the authorized CL **20** and countermeasures against the unauthorized AP **10**.

Embodiment 3

[0143] Hereafter, the points differing from the aforementioned Embodiments will mainly be described with reference to the drawings.

[0144] In Embodiment 1 and Embodiment 2, it is necessary to implement countermeasures for the damaged CLs **21** one by one, which requires a relatively long time to implement. Embodiment 3 shows a mode in which when there are many damaged CLs **21**, damage is mitigated by obstructing unauthorized communication more quickly.

Description of Configuration

[0145] FIG. **9** illustrates an example configuration of a monitoring device **100** according to Embodiment 3.

[0146] As shown in FIG. 9, a control unit **120** according to Embodiment 3, when compared to the control unit **120** according to Embodiment 1, further includes a deauthentication frame generation unit **128**.

[0147] The deauthentication frame generation unit **128** generates a deauthentication frame using an encryption key received from an unauthorized AP **10** when PMF is enabled in wireless communication between the unauthorized AP **10** and the damaged CL **21**, and sends communication data including the generated deauthentication frame to the damaged CL **21**. The encryption key is a common key in a wireless communication network to which the unauthorized AP **10** is participating.

Description of Operation

[0148] FIG. 10 is a flowchart showing an example of the operation of the monitoring device **100** according to Embodiment 3. The operation of the monitoring device **100** is described using FIG. 10.

(Step S301)

[0149] If the monitoring device **100** can successfully connect to the unauthorized AP **10**, the monitoring device **100** proceeds to step S302.

[0150] If the monitoring device **100** cannot successfully connect to the unauthorized AP **10**, the monitoring device **100** returns to step S109.

(Step S302)

[0151] The deauthentication frame generation unit **128** sends a deauthentication frame in which the source MAC address is set as the MAC address of the unauthorized AP **10**, to the wireless communication network in which the selected unauthorized AP is participating, by broadcasting within the LAN (Local Area Network) of the selected unauthorized AP a specified number of times. Broadcast communication in PMF is protected by BIP (Broadcast Integrity Protocol). Here, an IGTK (Integrity Group Transient Key), being an encryption key common within the LAN received from the AP, is used for integrity verification of the broadcast communication frame in BIP. Therefore, by the monitoring device **100** successfully connecting to the unauthorized AP **10** as a CL and receiving the IGTK, and using the received IGTK to send a broadcast deauthentication frame impersonating the unauthorized AP **10**, it is possible to disconnect the communication between the unauthorized AP **10** and the damaged CL **21** within the LAN in which the unauthorized AP **10** is participating. Setting the source MAC address to the MAC address of the unauthorized AP **10** corresponds to impersonating the unauthorized AP **10**.

(Step S303)

[0152] First, a communication monitoring unit **121** acquires communication frames from devices in a surrounding device group via a communication unit **110**.

[0153] Next, an information analysis unit **122** checks whether there is a communication frame from the selected unauthorized AP (i.e., a communication frame with the MAC address of the selected unauthorized AP as the source) among the communication frames acquired by the communication monitoring unit **121**.

[0154] In step S303, the communication monitoring unit **121** and the information analysis unit **122** appropriately execute at least part of step S101 and step S102.

(Step S304)

[0155] If there is a communication frame between the selected unauthorized AP and the damaged CL **21**, it suggests that the communication has not been properly disconnected, or that the damaged CL **21** has reconnected to the selected unauthorized AP after the communication was properly disconnected, so the monitoring device **100** returns to step S109.

[0156] If there is no communication frame from the selected unauthorized AP, the monitoring device **100** returns to step S101.

Description of Effects of Embodiment 3

[0157] As described above, according to this Embodiment, it is possible to disrupt unauthorized

communication between the damaged CL **21** and the unauthorized AP **10** using a deauthentication frame. Additionally, according to this Embodiment, it is possible to achieve effects similar to those of Embodiment 1.

Embodiment 4

[0158] Below, differences from the previously described Embodiment will mainly be described with reference to the drawings.

[0159] In Embodiment 3, there is a problem that if the unauthorized AP **10** has very high processing capability, a function as a disruption might not be achieved because the load on the unauthorized AP **10** cannot be sufficiently increased. Therefore, in Embodiment 4, a mode is shown where unauthorized communication is disrupted by having the damaged CL **21** restrain from communicating with the unauthorized AP **10**, instead of executing disruption on the unauthorized AP **10**.

Description of Configuration

[0160] FIG. **11** shows an example configuration of a monitoring device **100** according to Embodiment 4.

[0161] A control unit **120** according to Embodiment 4, as shown in FIG. **11**, is equipped with a probe response generation unit **126** instead of a communication connection unit **124** and a MAC address generation unit **125**.

[0162] The probe response generation unit **126** according to this Embodiment generates a beacon frame or a probe response frame.

[0163] As a specific example, the probe response generation unit **126** sends fourth transmission data to a damaged CL **21** to extend the time until the damaged CL **21** communicates next. The fourth transmission data is communication data including: a communication frame in which device identification information of an unauthorized AP is set as source device identification information so that the monitoring device **100** is recognized as the unauthorized AP **10**; and a communication frame in which either a QoS (Quality of Service) related field or a field related to communication control is set so that a waiting time taken until the damaged CL **21** sends a communication frame to the unauthorized AP **10** is not shorter than a first reference waiting time. The first reference waiting time may be defined in any manner.

[0164] The storage unit **130** according to Embodiment 4 stores a QoS parameter **136** instead of the used MAC address table **134**, as shown in FIG. **11**.

[0165] The QoS parameter **136** in Embodiment 4 is a value of the QoS related field in the IEEE (Institute of Electrical and Electronics Engineers) 802.11 frame format. As a specific example, the QoS parameter **136** expresses the EDCA (Enhanced Distributed Channel Access) parameter's CWmin (Contention Windows minimum; the minimum value of the communication waiting counter) or AIFSN (Arbitration InterFrame Space Number; the unit time of communication waiting time). The QoS parameter **136** is set to a large value in advance as a specific example. By sending a beacon frame or a probe response frame with the QoS parameter **136** being set to a large value to the damaged CL **21**, the waiting time until the damaged CL **21** sends a communication frame can be forcibly extended significantly. Setting the value of the QoS parameter **136** corresponds to tampering with the QoS parameter **136**.

[0166] Furthermore, depending on the implementation of the damaged CL **21**, a waiting time counter may be reset each time the damaged CL **21** receives a tampered beacon frame or probe response frame. In this case, it is possible to completely prevent the damaged CL **21** from communicating.

Description of Operation

[0167] FIG. **12** is a flowchart showing an example of the operation of the monitoring device **100** according to Embodiment 4. FIG. **12** is used to describe the operation of the monitoring device **100**.

(Step S401)

[0168] The probe response generation unit **126** refers to the damaged device information **133** and sends a probe response frame with the QoS parameter **136** appropriately being set, to each damaged CL **21** indicated by the damaged device information **133** via a communication unit **110**. At this time, the probe response generation unit **126** sets the transmission destination MAC address as the MAC address of each damaged CL **21**. The probe response generation unit **126** may send a broadcast beacon frame instead of a probe response frame.

Explanation of the Effects of Embodiment 4

[0169] As described above, according to this Embodiment, by using the QoS parameter **136** to restrain communication of each damaged CL **21**, it is possible to interfere with unauthorized communication between each damaged CL **21** and the unauthorized AP **10**. Additionally, according to this Embodiment, it is possible to achieve effects similar to those of Embodiment 1.

Other Configurations

<Modification 3>

[0170] Since Embodiment 4 is intended to extend the waiting time of the damaged CL **21**, there is a risk that the damaged CL **21** may communicate with the unauthorized AP **10** once the waiting time has passed. Therefore, in this modification, a mode is shown where by utilizing the RTS/CTS (Request To Send/Clear To Send) method, which is one method of communication control, the damaged CL **21** is restrained from communicating.

[0171] The RTS/CTS method is a method where the AP assigns a transmission right to the CL to control communication. In the RTS/CTS method, the CL sends an RTS frame to the AP, and the AP responds to the RTS by specifying one CL with a CTS frame, allowing only the specified CL to send data to the AP. By using this mechanism, the CL to which a transmission right is assigned in the CTS frame is tampered with to a CL that does not exist in the surrounding device group, and by sending the tampered CTS frame, all the damaged CLs **21** can be restrained from transmitting.

[0172] It should be noted since this modification assumes that the RTS/CTS method is being used, it cannot always be applied to the monitoring device **100**. However, many APs are equipped with a function to switch to the RTS/CTS method when communication quality deteriorates.

[0173] Below, the points different from Embodiment 4 are mainly described.

Description of Configuration

[0174] FIG. **13** shows an example configuration of a monitoring device **100** according to this modification.

[0175] A control unit **120** according to this modification, as shown in FIG. **13**, is equipped with a CTS generation unit **129** and a MAC address generation unit **125** instead of a probe response generation unit **126**.

[0176] The CTS generation unit **129** generates a CTS frame.

[0177] As a specific example, when the RTS/CTS method is used in a wireless communication network to which an unauthorized AP **10** and a damaged CL **21** are participating, the CTS generation unit **129** broadcasts data to the wireless communication network, the data including a communication frame in which device identification information different from device identification information of any device existing in the surrounding device group is set, as device identification information of a device to which a transmission right is to be assigned, in order to restrain the damaged CL **21** from communicating.

Description of Operation

[0178] FIG. **14** is a flowchart showing an example of the operation of the monitoring device **100** according to this modification. FIG. **14** is used to describe the operation of the monitoring device **100**.

(Step S421)

[0179] The MAC address generation unit **125** generates one MAC address. The MAC address generation unit **125** may generate a MAC address using random numbers or by incrementing the value of the MAC address indicated by the damaged device information **133**. The MAC address

generation unit **125** refers to the damaged device information **133** and checks to ensure that the generated MAC address is different from any MAC address of the damaged CL **21** indicated by the damaged device information **133**.

[0180] If the MAC address generation unit **125** confirms that the generated MAC address is different from any MAC address of the damaged CL **21** indicated by the damaged device information **133**, the MAC address generation unit **125** sends data indicating the generated MAC address to the CTS generation unit **129**. If the generated MAC address overlaps any MAC address of the damaged CL **21** indicated by the damaged device information **133**, the MAC address generation unit **125** regenerates a MAC address and reconfirms that the newly generated MAC address is different from any MAC address of the damaged CL **21** indicated by the damaged device information **133**.

(Step S422)

[0181] First, the CTS generation unit **129** specifies the MAC address indicated by the data received from the MAC address generation unit **125**, as the target of transmission right assignment.

[0182] Next, the CTS generation unit **129** generates a CTS frame with the transmission source MAC address being set as the MAC address of the unauthorized AP **10**, and sends the generated CTS frame via broadcast to the LAN of the unauthorized AP **10** through the communication unit **110**.

[0183] Additionally, the CTS generation unit **129** may unconditionally generate a CTS frame each time step S422 is executed and send the generated CTS frame each time the CTS frame is created. Furthermore, each time step S422 is executed, the communication monitoring unit **121** may reacquire the communication frames of the devices in the surrounding device group, and the CTS generation unit **129** may send the generated CTS frame only when the information analysis unit **122** confirms an RTS frame from the damaged CL **21** within the communication frames acquired by the communication monitoring unit **121**.

Explanation of the Effects of Modification 3

[0184] As described above, according to this modification, by using the RTS/CTS method to restrain the damaged CL **21** from communication, it is possible to disrupt unauthorized communication between the damaged CL **21** and the unauthorized AP **10**. Moreover, according to this modification, it is possible to achieve effects similar to those of Embodiment 1.

[0185] Additionally, in this modification, it is possible to restrain the damaged CL **21** from communication regardless of the waiting time of the damaged CL **21**. Therefore, according to this modification, it is possible to mitigate damage more effectively compared to Embodiment 4.

<Modification 4>

[0186] Embodiment 4 and Modification 3 are countermeasures against the unauthorized AP **10**. However, as shown in FIG. 17, by swapping the roles between the access point and the client, a monitoring device **100** can impersonate an authorized AP **30** and restrain an unauthorized CL **22** from communication. In other words, by appropriately adapting the configurations of Embodiment 4 and Modification 3, it is also possible to execute countermeasures against the unauthorized CL **22**.

[0187] Below, the case of adapting the configuration of Embodiment 4 will be described. It is assumed that an authorized AP **30** exists in the surrounding device group and that an unauthorized CL **22** is connected to the authorized AP **30**.

[0188] A probe response generation unit **126** sends fifth transmission data to the unauthorized CL **22** to restrain the unauthorized CL **22** from communication. The fifth transmission data is communication data including: a communication frame in which device identification information of the authorized AP **30** is set as the source device identification information so that the monitoring device **100** is recognized as the authorized AP **30**; and a communication frame in which either the QoS related field or the field related to communication control is set so that the waiting time taken until the unauthorized CL **22** sends the communication frame to the authorized AP **30** is not shorter

than a second reference waiting time. The second reference waiting time may be defined in any manner.

[0189] Below, the case of adapting the configuration of Modification 3 will be described. It is assumed that an authorized AP **30** exists in a surrounding device group and that an unauthorized CL **22** is connected to the authorized AP **30**.

[0190] When the authorized AP **30** and the unauthorized CL **22** are participating to a wireless communication network, the CTS generation unit **129**, broadcasts data to the wireless communication network, the data including a communication frame in which device identification information different from device identification information of any device existing in the surrounding device group is set as the device identification information of the device to which a transmission right is to be assigned, in order to restrain the unauthorized CL **22** from communicating.

[0191] Additionally, in Patent Literature 1, there is an issue where an unauthorized client that can no longer communicate due to the update of the encryption key can relatively easily reconnect to an authorized access point by requiring reconnection to the authorized access point using the same MAC address, as performed by the WIPS monitoring device. According to this modification, the unauthorized client is restrained from requiring reconnection to the authorized access point, making it not easy to reconnect to the authorized access point.

Embodiment 5

[0192] The following describes mainly the differences from the aforementioned Embodiments with reference to the drawings.

[0193] In Embodiment 2 and the modification of Embodiment 2, a method is adopted where the damaged CL **21** is once shifted to another channel to prevent communication with the damaged CL **21**, requiring time until the damaged CL **21** resumes authorized communication. Therefore, in Embodiment 5, to mitigate damage more quickly and resume authorized communication sooner, a mode is shown where the monitoring device **100** directly seizes the communication connection with the damaged CL **21** from the unauthorized AP **10**.

Description of Configuration

[0194] FIG. **15** shows a configuration example of a monitoring device **100** according to Embodiment 5.

[0195] The configuration of the monitoring device **100** is one in which the MAC address generation unit **125** and the used MAC address table **134** are excluded from the configuration of the monitoring device **100** according to Embodiment 1.

[0196] The communication connection unit **124** according to this Embodiment sends sixth transmission data to a damaged CL **21** to disconnect the communication connection between an unauthorized AP **10** and the damaged CL **21**. The sixth transmission data is communication data including: a communication frame in which the device identification information of the unauthorized AP **10** is set as the source device identification information so that the monitoring device **100** is recognized as the unauthorized AP **10**; a communication frame being set to disconnect communication connection between the unauthorized AP **10** and the damaged CL **21**; and a communication frame being set to establish communication connection with the damaged CL **21**.

Description of Operation

[0197] FIG. **16** is a flowchart showing an example of the operation of the monitoring device **100** according to Embodiment 5. FIG. **16** is used to describe the operation of the monitoring device **100**.

(Step S501)

[0198] The communication connection unit **124** refers to damaged device information **133** and selects one damaged CL **21** for which countermeasures have not been implemented. Hereinafter, the damaged CL **21** selected in step S501 is referred to as a third selected damaged CL.

(Step S502)

[0199] The communication connection unit **124** sends a probe response frame in which the sender is the unauthorized AP **10**'s MAC address and the transmission destination is the third selected damaged CL's MAC address, to the third selected damaged CL via a communication unit **110**, thereby executing the connection process with respect to the third selected damaged CL.

[0200] The communication connection unit **124**, after the connection process is successfully completed, sets a countermeasure implementation flag in the device information of the third selected damaged CL in the damaged device information **133**.

(Step S503)

[0201] If there is no device information in the damaged device information **133** where a countermeasure implementation flag is not set (that is, if a countermeasure implementation flag is set for all the damaged CLs **21** indicated by the damaged device information **133**), the monitoring device **100** proceeds to step S504. Otherwise, the monitoring device **100** proceeds to step S501.

(Step S504)

[0202] An unauthorized device judging unit **123** sets a countermeasure implementation flag in the device information of the selected unauthorized AP in an unauthorized device information **132**, assuming that countermeasures have been implemented for the selected unauthorized AP.

[0203] Additionally, since each damaged CL **21** might have been forced to download malware due to communicating with the selected unauthorized AP until immediately recently, there is a possibility that each damaged CL **21** might send an attack frame to an authorized AP **30**. Therefore, after the monitoring device **100** takes over the communication connection of each damaged CL **21**, the authorized AP **30** may monitor the communication of each damaged CL **21** for a certain period and implement countermeasures as necessary. Specific examples of countermeasures include displaying a warning screen on each damaged CL **21**, narrowing the communication bandwidth of each damaged CL **21**, or ignoring communication from each damaged CL **21**.

[0204] After the processing of step S504 is completed, the monitoring device **100** returns to step S101.

Explanation of the Effects of Embodiment 5

[0205] As described above, according to Embodiment 5, the monitoring device **100** can disconnect unauthorized communication between the unauthorized AP **10** and each damaged CL **21** by directly taking over the communication connection between the unauthorized AP **10** and each damaged CL **21**. Additionally, according to this Embodiment, the same effects as those of Embodiment 1 can be obtained.

[0206] Furthermore, according to Embodiment 5, since the monitoring device **100** provides authorized communication to each damaged CL **21**, it can provide authorized communication to each damaged CL **21** more quickly compared to Embodiment 2.

Other Configurations

<Modification 5>

[0207] Embodiment 5 is a countermeasure against an unauthorized AP **10**. However, as shown in FIG. 17, by swapping the roles between the access point and the client, the monitoring device **100** can also directly intercept communication from the authorized AP **30** to the unauthorized CL **22**. In other words, by appropriately adapting the configuration of Embodiment 5, it is also possible to execute countermeasures against the unauthorized CL **22**.

[0208] The following describes the adaptation of the configuration of Embodiment 5. It is assumed that an authorized AP **30** exists in a surrounding device group, and that an unauthorized CL **22** is connected to the authorized AP **30**.

[0209] A communication connection unit **124** sends seventh transmission data to the unauthorized CL **22** to disconnect the communication connection between the authorized AP **30** and the unauthorized CL **22**. The seventh transmission data is communication data including: a communication frame in which device identification information of the authorized AP **30** is set as

source device identification information so that a monitoring device **100** is recognized as the authorized AP **30**; a communication frame being set to disconnect communication connection between the authorized AP **30** and the unauthorized CL **22**; and a communication frame being set to establish communication connection with the unauthorized CL **22**.

[0210] Furthermore, according to this Modification, it is possible to solve the issues of the aforementioned Patent Literature 1.

Other Embodiments

[0211] It is possible to freely combine the previously described embodiments, modify any components of each embodiment, or omit any components in each embodiment.

[0212] Moreover, the embodiments are not limited to those shown as Embodiments 1 to 5, and various changes can be made as needed. The procedures described using flowcharts, etc., may be appropriately modified.

Reference Signs List

[0213] **10**: unauthorized AP; **20**: authorized CL; **21**: damaged CL; **22**: unauthorized CL; **30**: authorized AP; **51**: processor; **52**: memory; **53**: auxiliary storage device; **54**: input/output IF; **55**: communication device; **58**: processing circuit; **59**: signal line; **100**: monitoring device; **110**: communication unit; **120**: control unit; **121**: communication monitoring unit; **122**: information analysis unit; **123**: unauthorized device judging unit; **124**: communication connection unit; **125**: MAC address generation unit; **126**: probe response generation unit; **127**: channel selection unit; **128**: deauthentication frame generation unit; **129**: CTS generation unit; **130**: storage unit; **131**: communication frame information; **132**: unauthorized device information; **133**: damaged device information; **134**: used MAC address table; **135**: relation table; **136**: QoS parameter.

Claims

1. A monitoring device comprising processing circuitry to repeatedly execute connection to an unauthorized access point when the unauthorized access point exists within a surrounding device group, in order to fill simultaneous connection slots of the unauthorized access point, using each device identification information from a device identification information group consisting of one or more pieces of device identification information generated not to overlap each other; to connect to the unauthorized access point when the connection to the unauthorized access point fails, using device identification information of a damaged client being a communication device connected to the unauthorized access point, so as to disconnect communication connection between the unauthorized access point and the damaged client; and to connect to the unauthorized access point using device identification information different from any device identification information in the device identification information group, in order to fill the simultaneous connection slots of the unauthorized access point that have become available due to disconnection of the communication connection, the surrounding device group consisting of one or more devices that exist around the monitoring device and execute wireless communication, each device identification information in the device identification information group being different from the device identification information of the damaged client.
2. The monitoring device according to claim 1, wherein the processing circuitry judges whether the unauthorized access point exists within the surrounding device group based on a communication frame sent by each device in the surrounding device group.
3. The monitoring device according to claim 1, wherein each device identification information in the device identification information group is a MAC address, and the device identification information of the damaged client is a MAC address.
4. The monitoring device according to claim 1, wherein Protected Management Frames are enabled in wireless communication between the unauthorized access point and the damaged client.
5. The monitoring device according to claim 1 wherein the processing circuitry sends transmission

data to the damaged client so as to switch a channel of the damaged client to a first channel, the transmission data being communication data including: a communication frame in which device identification information of the unauthorized access point is set as source device identification information so that the monitoring device is recognized as the unauthorized access point; and a communication frame in which the first channel, being a channel where the unauthorized access point does not exist, is set as a channel switching destination.

6. The monitoring device according to claim 5, wherein when the monitoring device functions as an authorized access point, the processing circuitry sends transmission data to a damaged client whose channel has been switched to the first channel, so as to switch the channel of the damaged client to a second channel, the transmission data being communication data including: a communication frame in which the device identification information of the unauthorized access point is set as the source device identification information so that the monitoring device is recognized as the unauthorized access point; and a communication frame in which the second channel, being a channel where the unauthorized access point does not exist, is set as a channel switching destination, and wherein the processing circuitry switches a channel of the monitoring device to the second channel to establish authorized communication for the damaged client.

7. A monitoring method comprising by a computer being a monitoring device, repeatedly executing connection to an unauthorized access point when the unauthorized access point exists within a surrounding device group, in order to fill simultaneous connection slots of the unauthorized access point, using each device identification information from a device identification information group consisting of one or more pieces of device identification information generated not to overlap each other; connecting to the unauthorized access point when the connection to the unauthorized access point fails, using device identification information of a damaged client being a communication device connected to the unauthorized access point, so as to disconnect communication connection between the unauthorized access point and the damaged client; and connecting to the unauthorized access point using device identification information different from any device identification information in the device identification information group, in order to fill the simultaneous connection slots of the unauthorized access point that have become available due to disconnection of the communication connection, the surrounding device group consisting of one or more devices that exist around the monitoring device and execute wireless communication, each device identification information in the device identification information group being different from the device identification information of the damaged client.

8. A non-transitory computer readable medium recorded with a monitoring program which causes a monitoring device, being a computer, to execute a communication connection process comprising: repeatedly executing connection to an unauthorized access point when the unauthorized access point exists within a surrounding device group, in order to fill simultaneous connection slots of the unauthorized access point, using each device identification information from a device identification information group consisting of one or more pieces of device identification information generated not to overlap each other; connecting to the unauthorized access point when the connection to the unauthorized access point fails, using device identification information of a damaged client being a communication device connected to the unauthorized access point, so as to disconnect communication connection between the unauthorized access point and the damaged client; and connecting to the unauthorized access point using device identification information different from any device identification information in the device identification information group, in order to fill the simultaneous connection slots of the unauthorized access point that have become available due to disconnection of the communication connection, the surrounding device group consisting of one or more devices that exist around the monitoring device and execute wireless communication, each device identification information in the device identification information group being different from the device identification information of the damaged client.
